

HIROO, Japan

WMO: 474400

Lat: 42.2944N

Lon: 143.3167E

Elev: 112

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.7	12.7	-4.1	4.4	15.6	-0.8	5.3	17.2	22.6	30.6	20.1	30.4	5.2	270	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.5	81.9	71.0	78.0	68.6	74.2	66.4	72.7	79.7	70.6	75.9	68.6	72.4	9.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	111.9	76.4	68.6	105.5	74.0	67.0	99.8	71.6	36.6	80.0	34.7	75.8	33.0	72.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	17.2	14.5	DB	4.6	88.8	3.3	3.5	2.2	91.3	0.3	93.4	-1.6	95.4	-4.0	97.9	(4)
			WB	3.4	75.6	3.3	1.7	1.0	76.9	-0.9	77.9	-2.8	78.8	-5.2	80.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	23.6	24.2	31.9	41.5	50.1	55.3	62.6	66.2	62.2	52.1	40.9	29.4
	DBStd	15.78	5.45	5.73	6.06	6.34	6.94	5.84	5.79	5.53	5.25	5.47	6.75	5.93
	HDD50	3428	819	721	562	267	86	13	0	0	0	40	280	640
	HDD65	7431	1284	1141	1027	705	464	299	117	50	117	400	722	1105
	CDD50	1646	0	0	0	13	89	170	391	502	367	105	9	0
	CDD65	174	0	0	0	1	2	7	43	87	33	1	0	0
	CDH74	859	0	0	0	2	32	51	260	391	122	1	0	0
	CDH80	192	0	0	0	0	6	8	61	101	16	0	0	0
Wind	WSAvg	6.3	6.8	6.8	7.2	7.1	6.4	5.0	4.5	4.8	5.6	6.5	7.2	7.1
Precipitation	PrecAvg	61.6	2.6	2.2	3.0	4.2	5.9	5.2	6.5	7.9	9.5	6.3	4.9	3.5
	PrecMax	82.1	6.0	5.5	5.1	14.5	11.2	15.7	15.3	19.1	19.0	13.5	12.1	8.9
	PrecMin	42.4	0.4	0.4	0.7	0.6	2.0	1.0	1.0	2.1	2.5	1.9	1.3	1.1
	PrecStd	10.6	1.7	1.5	1.5	3.0	2.8	3.7	3.7	4.2	4.2	3.4	2.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	45.0	53.5	69.3	77.8	80.4	86.0	88.1	82.1	71.2	62.5	51.6
		MCWB	35.7	38.9	43.9	52.0	58.0	64.1	72.9	74.7	70.4	60.7	53.7	44.9
	2%	DB	38.2	40.5	48.2	61.9	70.9	73.7	80.9	82.5	77.5	67.0	58.8	45.7
		MCWB	33.1	34.6	40.6	48.7	55.6	61.9	71.1	72.4	68.4	57.7	50.9	40.3
	5%	DB	35.1	37.0	44.4	56.6	65.9	68.7	76.2	78.4	73.7	63.8	55.7	42.1
		MCWB	30.3	31.8	37.5	46.3	54.0	59.7	68.6	70.5	66.3	56.5	49.2	36.9
	10%	DB	32.8	34.1	41.3	52.0	61.4	63.6	70.8	74.0	70.3	61.3	52.5	38.9
		MCWB	28.7	29.8	35.2	43.3	52.1	58.2	66.0	68.5	64.7	55.1	46.8	33.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	40.0	45.9	54.2	60.7	66.6	74.4	76.0	73.4	63.7	56.0	46.0
		MCDB	40.8	43.5	52.2	64.4	71.6	76.6	83.5	85.8	78.7	67.3	60.3	50.5
	2%	WB	33.9	35.3	41.4	50.1	57.7	63.3	71.9	73.4	70.5	60.3	52.6	41.3
		MCDB	37.1	39.4	47.4	60.4	67.7	70.6	79.7	80.6	75.0	64.5	57.2	44.5
	5%	WB	31.4	32.3	38.4	47.1	55.3	60.8	69.2	71.3	68.1	58.1	50.0	37.5
		MCDB	34.4	36.0	43.5	54.8	63.8	66.4	74.8	76.7	71.6	62.1	54.6	41.3
	10%	WB	28.9	30.3	35.9	44.5	52.8	58.5	66.4	69.4	65.7	56.2	47.0	34.5
		MCDB	32.5	33.8	40.3	50.7	59.7	62.7	70.0	73.2	69.0	60.2	51.7	38.4
Mean Daily Temperature Range	5% DB	MDBR	14.2	14.2	13.2	14.3	13.7	10.5	9.4	9.5	11.5	14.2	14.1	13.4
		MCDBR	15.6	16.9	17.4	23.1	23.9	21.0	19.4	17.9	16.6	16.8	18.4	16.1
		MCWBR	12.8	13.5	12.4	13.4	12.4	10.9	10.4	9.5	9.7	11.6	14.6	13.0
	5% WB	MCDBR	15.1	16.3	16.6	20.5	20.9	17.0	17.7	15.4	13.6	14.3	17.4	15.8
		MCWBR	13.3	13.7	12.7	12.9	12.1	9.8	10.4	8.9	8.9	11.1	14.9	13.6
Clear-Sky Solar Irradiance	taub	0.335	0.385	0.466	0.523	0.542	0.511	0.499	0.480	0.422	0.404	0.383	0.352	
	taud	2.167	2.011	1.866	1.811	1.855	2.022	2.128	2.197	2.316	2.225	2.183	2.156	
	Ebn at Noon	246	249	242	239	239	247	248	249	256	243	226	226	
	Edh at Noon	31	43	57	65	64	54	48	44	36	35	31	30	
All-Sky Solar Radiation	RadAvg	607	878	1230	1501	1572	1539	1371	1286	1145	934	625	497	
	RadStd	33	58	94	132	148	133	134	118	101	60	44	40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.56	N/A	N/A	N/A	N/A	+1.49	N/A	-175	+131	N/A
Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-113	N/A	N/A

Nomenclature: See separate page